The Graffiti.pc conjecture $f(G) \geq \lceil (p(G) + b(G) + 1)/2 \rceil$: an equivalent deficiency form, finite reductions, and computational verification

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Abstract

Let G be a connected graph on $n \geq 2$ vertices. Write $f(G)$ for the maximum order of an induced forest, $b(G)$ for the maximum order of an induced bipartite subgraph, and $p(G)$ for the minimum number of vertex-disjoint paths covering $V(G)$. Conjecture 40 of the *Written on the Wall II* (Graffiti.pc) collection asserts $f(G) \geq \lceil (p(G) + b(G) + 1)/2 \rceil$. Introducing the complementary parameters $\tau = n - f$ (decycling number), $o = n - b$ (odd-cycle transversal number), and $\ell = n - p$ (number of edges in a maximum spanning linear forest), we prove the conjecture is equivalent to the linear inequality $\ell + o \geq 2\tau + 1$. For bipartite G this reads $\ell \geq 2\tau + 1$; an incidence-selection flow model reduces this case to two explicitly stated open lemmas on forced cores and cycle-erasing descent. We prove the surrounding structure and a carrier–cycle separation lemma. On the general side we prove, conditional on the bipartite case, $\ell + 2o \geq 2\tau + 1$. We also show that the single-vertex reduction property SV, left open in version 1, is false: its minimum counterexamples have order 11, including a graph with a unique minimum odd-cycle transversal whose deletion lowers the path-cover number. The associated path-cover lifting property also fails. The alternative lifting inequality $\ell(G) \geq 2\tau(G - T) + |T| + 1$ survives exhaustive verification through $n = 10$ (even for every minimum T); we prove it whenever $p(G) \leq 2c_e(G - T) - 1$. Finally, under frozen and independently adversarially reverified protocols, the original conjecture holds for every connected graph on at most 11 vertices, Lemma A has no counterexample in 2,391,365,507 nontrivial instances through $n = 14$, and the stronger fixed-edge selection property F2 holds through $n = 13$.

1 Introduction

For a finite simple graph $G = (V, E)$ on $n = |V|$ vertices, let

$$f(G) = \text{max. order of an induced forest}, \quad b(G) = \text{max. order of an induced bipartite subgraph},$$

and let $p(G)$ be the *path cover number*: the minimum number of vertex-disjoint simple paths (isolated vertices allowed as trivial paths) whose vertex sets partition V. The following appears as Conjecture 40 in DeLaviña's *Written on the Wall II* (WOWII), the conjecture stream of the Graffiti.pc system.

Conjecture 1.1 (WOWII #40)**.** *For every connected graph G on $n \geq 2$ vertices,*

$$f(G) \;\geq\; \left\lceil \frac{p(G) + b(G) + 1}{2} \right\rceil.$$

The specialization $p(G) = 1$ (graphs with a Hamiltonian path) was highlighted early: there the bound reads $f \geq \lceil (b+2)/2 \rceil = \lfloor b/2 \rfloor + 1$. The conjecture is recorded as open on the maintained WOWII list (last updated 2026-07-23) and is carried, classified research-open, in the Lean formalization repository `google-deepmind/formal-conjectures` as `GraphConjecture40`. A search of the standard combinatorics literature at the time of writing located no proof, counterexample, or implying theorem (MathSciNet and zbMATH were not searched; this is a search report, not a nonexistence claim). The closest related work is the Hall-type path-cover condition of Lavrov and Vandenbussche [1] and the adjacent super-cyclic literature; see Section 7.

Results. This paper contributes four things.

(i) **An equivalent deficiency form** (Section 2). With $\tau = n - f$, $o = n - b$, $\ell = n - p$, Conjecture 1.1 is equivalent to $\ell(G) + o(G) \geq 2\tau(G) + 1$. We give the exact equality analysis, including a closed margin formula $\mu = n_r - n_{r-1}$ for traceable complete multipartite graphs.

(ii) **A reduction of the bipartite case** (Sections 3–4). For bipartite G the target is $\ell \geq 2\tau + 1$. Through an incidence-selection model we prove a generalized Hall criterion, a rooted forest extension bound, a repaired residual-augmentation flow theorem, a descent argument, and a first carrier-family lemma. The two general lemmas remain open; computationally, Lemma A survives through $n = 14$ and the direct property F2 through $n = 13$.

(iii) **The general-case gap and its surviving route** (Section 5). Conditional on the bipartite case we prove the coefficient-two bound $\ell + 2o \geq 2\tau + 1$. The single-vertex property SV is false, with minimum counterexamples at $n = 11$, so its implication theorem has a false hypothesis. The path-cover lifting property fails with it. The alternative lifting inequality survives, including its universal minimum-OCT form, and we prove it on a structured class containing every tight instance through $n = 8$.

(iv) **Exhaustive verification to** $n = 11$ (Section 6). Under a preregistered, hash-frozen, dual-engine protocol, Conjecture 1.1 holds for every connected graph on ≤ 11 vertices (1,006,700,565 graphs at $n = 11$), with zero violations. The new lemma-level and lifting computations were independently rerun from the paper's definitions under an adversarial verification protocol.

Two lemmas in the bipartite reduction remain open; the general-to-bipartite step at coefficient one remains open. The paper states exactly what is proved, what is conditional, and what is verified only computationally.

2 The deficiency form

Throughout, G is connected with $n \geq 2$ vertices. Define the three *deficiency parameters*

$$\tau(G) = n - f(G), \qquad o(G) = n - b(G), \qquad \ell(G) = n - p(G).$$

Here τ is the *decycling number* (minimum size of a feedback vertex set: deleting a maximum induced forest leaves a minimum FVS), o is the *odd-cycle transversal number* (deleting a maximum induced bipartite subgraph leaves a minimum odd-cycle transversal), and ℓ is the maximum number of edges in a spanning *linear forest* (a disjoint union of paths): a path cover by p paths has exactly $n - p$ edges, so maximizing edges is minimizing paths.

Theorem 2.1 (Equivalence)**.** *For every connected graph G on $n \geq 2$ vertices,*

$$f(G) \geq \left\lceil \frac{p(G) + b(G) + 1}{2} \right\rceil \quad \Longleftrightarrow \quad \ell(G) + o(G) \;\geq\; 2\tau(G) + 1.$$

In particular, for bipartite G (where $o = 0$) the conjecture is $\ell(G) \geq 2\tau(G) + 1$.

Proof. The ceiling inequality $f \geq \lceil (p+b+1)/2 \rceil$ is equivalent, over the integers, to $2f \geq p+b+1$: indeed $f \geq \lceil m/2 \rceil$ iff $2f \geq m$ for any integer m, since $2f$ is even and $\lceil m/2 \rceil$ is the least integer t with $2t \geq m$. Substitute $f = n - \tau$, $b = n - o$, $p = n - \ell$:

$$2(n-\tau) \;\geq\; (n-\ell) + (n-o) + 1 \iff -2\tau \geq -\ell - o + 1 \iff \ell + o \;\geq\; 2\tau + 1. \qquad \square$$

We verified Theorem 2.1 numerically on all 995 connected graphs with $2 \leq n \leq 7$ (checker `phase0.c`, Section 6): the two sides agreed in every case.

Margin and tightness. Define the *linear margin*

$$\mu(G) \;:=\; \ell(G) + o(G) - \big(2\tau(G) + 1\big).$$

Conjecture 1.1 is the assertion $\mu(G) \geq 0$. We call G *linear-tight* when $\mu(G) = 0$. One must not conflate linear tightness with equality in the original ceiling statement: because of the parity of the ceiling, equality $f = \lceil (p+b+1)/2 \rceil$ holds precisely when $\mu(G) \in \{0, 1\}$. Thus "ceiling-equality" is the union of the linear-tight shell $\mu = 0$ and the neighboring shell $\mu = 1$.

Proposition 2.2 (Equality families)**.** *Let $G = K_{n_1,\ldots,n_r}$ be complete multipartite with $r \geq 2$ parts ordered $n_1 \leq \cdots \leq n_r$ and $n = \sum_i n_i$. Then*

$$p(G) = \max(1,\, 2n_r - n), \qquad \mu(G) = \begin{cases} n_r - n_{r-1}, & p = 1 \quad (\textit{i.e. } 2n_r \leq n+1), \\ (n - n_r) - n_{r-1} + 1, & p > 1 \quad (\textit{i.e. } 2n_r > n+1). \end{cases}$$

Consequently G is linear-tight ($\mu = 0$) iff it is traceable and its two largest parts are equal ($n_{r-1} = n_r$); this includes K_n and $K_{a,a}$. And $\mu = 1$ for every $K_{a,b}$ with $a < b$ and for every star $K_{1,m}$ with $m \geq 2$. Separately, C_3, C_4, and K_2 are linear-tight.

Proof. An induced subgraph of $K_{n_1,\ldots,n_r}$ is again complete multipartite, hence a forest iff it is edgeless or a star; the largest induced forest is one vertex plus a largest part, so $f = n_r + 1$ and $\tau = n - n_r - 1$. An induced subgraph is bipartite iff at most two parts are nonempty, so $b = n_r + n_{r-1}$ and $o = n - n_r - n_{r-1}$. A complete multipartite graph is traceable iff the largest part does not exceed the rest by more than one, i.e. $2n_r \leq n+1$; otherwise the $2n_r - n$ excess vertices of the largest part each start a new path, so $p = \max(1, 2n_r - n)$.

If $p = 1$ then $\ell = n - 1$ and

$$\mu = (n-1) + (n - n_r - n_{r-1}) - \big(2(n - n_r - 1) + 1\big) = n_r - n_{r-1}.$$

If $p > 1$ then $\ell = n - p = n - (2n_r - n) = 2(n - n_r)$ and

$$\mu = 2(n-n_r) + (n - n_r - n_{r-1}) - \big(2(n - n_r - 1) + 1\big) = (n - n_r) - n_{r-1} + 1.$$

The two formulas agree on the boundary $2n_r = n+1$. For $K_{a,b}$ with $a < b$: if $b = a+1$ (traceable) the first formula gives $\mu = b - a = 1$; if $b \geq a+2$ the second gives $\mu = (a+b-b) - a + 1 = 1$;

so every $K_{a,b}$ with $a < b$ has $\mu = 1$. For a star $K_{1,m}$ ($m \geq 2$), $n_r = m, n_{r-1} = 1, n = m + 1$, and the $p > 1$ formula gives $\mu = (m + 1 - m) - 1 + 1 = 1$ (matching $p = m - 1$). For linear tightness: in the $p > 1$ regime $(n - n_r) - n_{r-1} + 1 \geq 1$ because $n_{r-1} \leq n - n_r$ (the second-largest part is at most the sum of all non-largest parts), so $\mu = 0$ forces $p = 1$ and then $n_{r-1} = n_r$; conversely $p = 1$ with $n_{r-1} = n_r$ gives $\mu = 0$. Finally direct evaluation gives $\mu = 0$ for K_2 ($\ell = 1, o = 0, \tau = 0$), C_3 $(2, 1, 1)$, and C_4 $(3, 0, 1)$. □

These formulas locate the extremal graphs of the deficiency form: the traceable complete multipartite graphs with equal top two parts are exactly linear-tight, and the whole complete bipartite family $K_{a,b}$ ($a \neq b$) together with the stars sits on the neighboring $\mu = 1$ shell. They cross-check against the gold invariants of Section 6: $K_{10}\,(f, b, p) = (2, 2, 1)$, $K_{5,5} = (6, 10, 1)$, $K_{1,9} = (10, 10, 8)$, $C_{10} = (9, 10, 1)$.

3 The bipartite case: reduction to two lemmas

Fix a connected nontrivial bipartite graph G with bipartition $V = L \cup R$. The target is $\ell(G) \geq 2\tau(G) + 1$. If $\tau(G) = 0$ then G is a nontrivial forest, has an edge, and that edge (with all other vertices as trivial paths) is a spanning linear forest of $\geq 1 = 2\tau + 1$ edges; so assume $\tau \geq 1$.

3.1 The incidence-selection model

Let S be a minimum feedback vertex set, $|S| = \tau$. Then $H := G - S$ is a forest (deleting an FVS destroys all cycles). For a vertex set X we write $G[H \cup X]$ for the subgraph induced on $H \cup X$. We build a capacitated selection problem on the S–H incidences.

Definition 3.1 (Selection model)**.** Fix an edge $e = (x, y)$ of H (its endpoints lie in opposite bipartition classes). Each $s \in S$ has *demand* 2. Each $h \in H$ has *capacity* $c(h) = 2$, except $c(x) = c(y) = 1$. A *feasible selection* J assigns to each $s \in S$ two distinct S–H incidences at s so that each $h \in H$ receives at most $c(h)$ selected incidences. A *selected cycle* of J is a cyclic component of the incidence graph J.

The point of the model is the following counting fact, proved in Section 3.5: an *acyclic* feasible selection J (with $J \cup \{e\}$ acyclic) yields a spanning linear forest of G with exactly $2|S| + 1$ edges, hence $\ell \geq 2\tau + 1$. So the bipartite target reduces to the existence of an acyclic feasible selection for some fixed edge e.

We record the canonical semantics of a cycle-erasing move once, and use it throughout.

Definition 3.2 (CES move)**.** A toggle from a feasible selection J to a feasible selection J' is *cycle-erasing (CES)* for a designated selected cycle C of J when (i) C is not a selected cycle of J', and (ii) every selected cycle of J' was already a selected cycle of J. This is a semantic condition (compare the selected-cycle sets before and after); it implies a strict decrease in the number of selected cycles. CES is *not* defined merely as "fewer selected cycles"; a strict decrease alone would already suffice for termination of the descent (Section 3.5), but the stronger no-new-cycle form is what the mechanism produces.

3.2 Hereditary tightness and the Hall criterion

Lemma 3.3 (Hereditary FVS tightness)**.** *For every $X \subseteq S$, $\tau(G[H \cup X]) = |X|$.*

Proof. X is an FVS of $G[H \cup X]$ (deleting it leaves H, a forest), so $\tau(G[H \cup X]) \le |X|$. Conversely, suppose D is an FVS of $G[H \cup X]$ with $|D| < |X|$. Then $(S \setminus X) \cup D$ is an FVS of G: deleting it leaves a subgraph of $G[H \cup X] - D$, which is a forest. Its size is $|S| - |X| + |D| < |S|$, contradicting minimality of S. Hence $\tau(G[H \cup X]) = |X|$. □

Encode feasibility as a flow. Build a network N with a source σ; arcs $\sigma \to s$ of capacity 2 for each $s \in S$; an arc $s \to h$ of capacity 1 for each incidence; and arcs $h \to$ sink of capacity $c(h)$. An integral flow saturating every $\sigma \to s$ is exactly a feasible selection. For $X \subseteq S$ put

$$\rho(X) = \sum_{h \in H} \min\big(c(h), \deg_X(h)\big), \qquad \delta(X) = \rho(X) - 2|X|,$$

where $\deg_X(h)$ is the number of neighbours of h in X.

Lemma 3.4 (Generalized Hall criterion)**.** *A feasible selection exists iff $\rho(X) \ge 2|X|$ for every $X \subseteq S$.*

Proof. Consider any σ–sink cut (A, B) with $\sigma \in A$, sink $\in B$, and set $X = S \cap A$. The forward capacity is: source arcs $\sigma \to s$ for $s \in S \cap B$ cost 2 each, total $2(|S| - |X|)$; an incidence arc $s \to h$ with $s \in A, h \in B$ can be forward only for $s \in X$, cost 1; a sink arc $h \to$ sink with $h \in A$ costs $c(h)$. For each h, its contribution is minimized by placing it on the cheaper side, giving $\min(c(h), \deg_X(h))$. Hence the cheapest cut with a given X has capacity $2(|S| - |X|) + \rho(X) = 2|S| + \delta(X)$, and the global minimum cut is $2|S| + \min_X \delta(X)$. By max-flow/min-cut the maximum flow equals $\min(2|S|, 2|S| + \min_X \delta(X))$, which is $2|S|$ (a feasible selection) iff $\min_X \delta(X) \ge 0$. □

An incidence $a = (s, h)$ is *forced* if it belongs to every feasible selection; the *forced core* is the graph of all forced incidences.

Lemma 3.5 (Forced $\iff$ tight-set exposed, F1)**.** *Fix a feasible selection. An incidence $a = (s, h)$ is forced iff some ρ-tight set X (i.e. $\delta(X) = 0$) exposes it: $s \in X$ and $\deg_X(h) \le c(h)$.*

Proof. Deleting the arc a from N lowers $\rho(X)$ by exactly one iff $s \in X$ and $\deg_X(h) \le c(h)$. If a is forced then $N - a$ has no feasible flow, so by Lemma 3.4 some X has $\rho_{N-a}(X) < 2|X|$; since $\rho(X) \ge 2|X|$ held, X was tight and a lowered it, so X exposes a. Conversely, if a tight X exposes a, then in $N - a$ we have $\rho_{N-a}(X) = 2|X| - 1 < 2|X|$, so no feasible selection omits a. □

Lemma 3.6 (Tight-set lattice)**.** *δ is submodular and nonnegative on subsets of S; hence if X, Y are tight so are $X \cap Y$ and $X \cup Y$.*

Proof. Each map $X \mapsto \min(c(h), \deg_X(h))$ is submodular, being the composition of the concave nondecreasing function $t \mapsto \min(c(h), t)$ with the modular map $X \mapsto \deg_X(h)$; hence ρ and $\delta(X) = \rho(X) - 2|X|$ are submodular, and feasibility gives $\delta \ge 0$. If $\delta(X) = \delta(Y) = 0$ then $0 \le \delta(X \cap Y) + \delta(X \cup Y) \le \delta(X) + \delta(Y) = 0$, forcing both to vanish. □

3.3 Feasibility for every fixed edge

We show a feasible selection actually exists—for *every* choice of the distinguished edge e. This uses a rooted forest bound of independent interest.

Lemma 3.7 (Rooted forest extension)**.** *Let G be bipartite, H an induced forest, and X an independent set contained in one bipartition class. Fix an edge $e = ab$ of H; give a, b capacity 1 and every other H-vertex capacity 2, and put $\rho_e(X) = \sum_h \min(c_e(h), \deg_X(h))$. Then*

$$\tau(G[H \cup X]) \;\le\; \lfloor \rho_e(X)/2 \rfloor .$$

Proof. Let r be the endpoint of e in the bipartition class opposite X; the other endpoint lies in X's class and so has no X-neighbour. Thus r is the only e-endpoint that can have X-neighbours. Induct on $k = \#\{h \in H \setminus \{r\} : \deg_X(h) \geq 2\}$.

Step ($k > 0$). Pick $h \in H \setminus \{r\}$ with $\deg_X(h) \geq 2$; then h is neither endpoint of e, so $c_e(h) = 2$. Put $K = G[H \cup X]$, $K' = K - h$. Then $H - h$ is a forest containing e, so the lemma applies to K', and deleting h lowers ρ_e by $\min(2, \deg_X(h)) = 2$. If D is an FVS of K' then $D \cup \{h\}$ is an FVS of K, so $\tau(K) \leq 1 + \lfloor(\rho_e^K(X) - 2)/2\rfloor = \lfloor \rho_e^K(X)/2\rfloor$.

Base ($k = 0$). Every $h \in H \setminus \{r\}$ has ≤ 1 X-neighbour. *Case 1* ($\deg_X(r) \leq 1$): every h has ≤ 1 X-neighbour, so $\rho_e(X)$ equals the number of X–H incidences. Let $D = \{x \in X : \deg_H(x) \geq 2\}$; each contributes ≥ 2 incidences, so $2|D| \leq \rho_e(X)$. In $K - D$ every surviving X-vertex has degree ≤ 1; adjoining leaves to the forest H keeps it a forest, so $K - D$ is a forest and $\tau(K) \leq |D| \leq \lfloor \rho_e(X)/2\rfloor$. *Case 2* ($\deg_X(r) \geq 2$): let m be the number of X-to-$(H \setminus \{r\})$ incidences; each non-root H-vertex contributes ≤ 1 and r contributes $c_e(r) = 1$, so $\rho_e(X) = m + 1$. Let $Q = \{x \in X : \deg_{H-r}(x) \geq 2\}$, $q = |Q|$, so $2q \leq m$. If $m \geq 2q + 1$, take $D = Q \cup \{r\}$: after deleting r, each $x \notin Q$ has ≤ 1 remaining H-neighbour, so $K - D$ is a forest and $|D| = q + 1 \leq \lfloor(m + 1)/2\rfloor = \lfloor \rho_e(X)/2\rfloor$. If $m = 2q$, take $D = Q$: every $x \in Q$ has exactly two non-root incidences and every $x \notin Q$ none, so surviving X-vertices have degree ≤ 1 (only possibly to r) and $K - D$ is a forest, with $|D| = q = \lfloor(2q + 1)/2\rfloor = \lfloor \rho_e(X)/2\rfloor$. □

Corollary 3.8 (Initial feasibility for every e)**.** *For every $X \subseteq S$ and every fixed edge $e \in E(H)$, $\rho_e(X) \geq 2|X|$; hence a feasible selection exists for every distinguished edge e.*

Proof. Split X into $X_L = X \cap L$ and $X_R = X \cap R$, each independent. Every $h \in H \cap L$ has neighbours only in X_R and every $h \in H \cap R$ only in X_L, so the supports are disjoint and $\rho_e(X) = \rho_e(X_L) + \rho_e(X_R)$. If $\rho_e(X) < 2|X|$ then one side, say X_L, has $\rho_e(X_L) < 2|X_L|$; then Lemma 3.7 gives $\tau(G[H \cup X_L]) \leq \lfloor \rho_e(X_L)/2\rfloor < |X_L|$, contradicting hereditary tightness (Lemma 3.3). Thus $\rho_e(X) \geq 2|X|$ for all $X \subseteq S$, and Lemma 3.4 gives feasibility. □

3.4 Parity and residual augmentation

Lemma 3.9 (Fixed-edge parity)**.** *The fixed edge $e = (x, y)$ lies on no selected cycle, and $J \cup \{e\}$ is acyclic iff J is.*

Proof. If $J \cup \{e\}$ had a cycle through e, deleting e leaves an x–y path of S–H incidence edges, which alternates $H, S, H, \ldots$ and so has even length. But x, y are adjacent in bipartite G, hence in opposite classes, so every x–y path has odd length — a contradiction. □

The next theorem is the corrected engine of the descent. (An earlier symmetric-difference sketch was superseded; the argument below is a direct max-flow augmenting-path proof.)

Theorem 3.10 (Residual augmentation)**.** *Let J be a feasible selection and $a = (s, h)$ a selected incidence. Delete the arc a from N entirely; call the result $N - a$. Then a feasible selection omitting a exists iff $N - a$ admits a flow of value $2|S|$ iff there is a simple residual σ–sink augmenting path in $N - a$. Consequently a is avoidable iff it is avoidable by a single residual augmentation.*

Proof. J is a value-$2|S|$ integral flow in N. Form $J - a$ by deleting one complete unit-flow path $\sigma \to s \to h \to$ sink (the unit routed through a): the flow on $\sigma \to s$ drops from 2 to 1; the arc a is absent from $N - a$; the flow on $h \to$ sink drops by 1; all other arc flows are unchanged. Conservation is preserved at s (inflow and outflow both drop by 1) and at h (likewise); every other vertex is untouched. So $J - a$ is a valid integral flow of value $2|S| - 1$ in $N - a$, in which $\sigma \to s$ is the only unsaturated source arc.

A feasible selection omitting a is a value-$2|S|$ flow in $N - a$ (it saturates every $\sigma \to s$ and never uses a). Conversely, by the augmenting-path theorem applied to $J - a$, the network $N - a$ admits a larger (value-$2|S|$) flow iff there is a simple residual σ–sink path from the deficient s; augmenting along it saturates $\sigma \to s$ and yields a feasible selection. Its incidence portion (delete the source/sink arcs) is exactly an alternating augmenting path from s to an H-vertex of residual capacity. The released arc a is absent from $N - a$, so no residual path can re-add it. □

Corollary 3.11. *There is no incidence that is globally avoidable yet admits no single residual augmentation avoiding it; "avoidable" and "one-move avoidable" coincide.*

3.5 Descent, and the reduction to two lemmas

Lemma 3.12 (Descent)**.** *If, from every feasible selection with a selected cycle, some CES toggle exists, then repeatedly applying CES toggles terminates in an acyclic feasible selection.*

Proof. A CES toggle destroys the target cycle and creates none, so it strictly lowers the (nonnegative integer) number of selected cycles. A strictly decreasing sequence of nonnegative integers terminates; the terminal selection is acyclic. □

We now name the two properties that a CES toggle needs, and package the whole bipartite case. Write $F2$ for the statement that an acyclic feasible selection exists for some fixed edge e.

Theorem 3.13 (Reduction of the bipartite case)**.** *Suppose the following two lemmas hold for all connected nontrivial bipartite graphs.*

- ***A.*** *(Forced core acyclic.) In the model of Definition 3.1, the forced core contains no cycle.*
- ***B.*** *(Cycle-erasing descent.) Every selected cycle that contains an avoidable (non-forced) incidence admits a CES toggle.*

Then every connected nontrivial bipartite graph satisfies $\ell \geq 2\tau + 1$, i.e. Conjecture 1.1 holds in the bipartite case.

Proof. Take $\tau \geq 1$ (the $\tau = 0$ case was handled at the start of Section 3). We first check $E(H) \neq \varnothing$, so a fixed edge exists. If $E(H) = \varnothing$, take $X = \{s\}$ for any $s \in S$: then $G[H \cup \{s\}]$ has edges only at s (a star forest), so $\tau(G[H \cup \{s\}]) = 0$, contradicting hereditary tightness (Lemma 3.3), which gives 1. So fix $e \in E(H)$.

By Corollary 3.8 a feasible selection exists. By Lemma A the forced core is acyclic, so every selected cycle contains an avoidable incidence: forced incidences belong to every feasible selection, so a selected cycle all of whose incidences were forced would be a cycle of the forced core, contradicting acyclicity. Hence Lemma B applies to every selected cycle and supplies a CES toggle, and the descent (Lemma 3.12) terminates in a feasible selection J with no selected cycle. By Lemma 3.9, $J \cup \{e\}$ is acyclic. Now $J \cup \{e\}$ has maximum degree 2: every $s \in S$ has selected degree 2; every $h \in H$ has degree $\leq c(h) \leq 2$; and x, y keep degree ≤ 2 because $c(x) = c(y) = 1$ leaves one unit for the edge e. An acyclic graph of maximum degree 2 is a disjoint union of paths, i.e. a linear forest. It has $2|S|$ incidence edges plus the edge e, and it spans $S \cup H = V$; every s has degree 2, every internal vertex degree ≤ 2. Hence it is a spanning linear forest with $2|S| + 1$ edges, so $\ell \geq 2|S| + 1 = 2\tau + 1$. □

Remark 3.14. Theorem 3.13 isolates the open content precisely. All of Sections 3.2–3.5 — the Hall criterion, the F1 characterization, the tight-set lattice, the rooted forest bound and feasibility for every e, fixed-edge parity, the residual-augmentation flow theorem, and the descent — are proved unconditionally. What remains for the bipartite case is exactly Lemma A and Lemma B.

4 The two open lemmas

We state the two lemmas in the strongest proved-adjacent forms and record the footholds that constrain any proof.

4.1 Lemma A: the forced core is a forest

It is essential that Lemma A use hereditary tightness, not merely Hall feasibility. The abstract even-cycle system (S_i adjacent to H_i, H_{i+1}, demand 2, capacity 2) is Hall-feasible yet its unique feasible selection is the whole cycle, so its forced core is a cycle; there $\tau(G[H \cup S]) = 1 < |S|$, i.e. hereditary tightness fails. Thus Lemma A is exactly the implication

$$\text{hereditary FVS tightness for all } X \subseteq S \implies \text{forced core acyclic},$$

and this implication is open in general. We state the lemma without a connectedness hypothesis, since none of the footholds uses it and the disconnected form implies the connected one.

Open Problem 4.1 (Lemma A°)**.** Let G be a finite simple bipartite graph, *not necessarily connected*, S a minimum FVS, $H = G - S$, and e a fixed edge of H with capacities as in Definition 3.1. Prove that the forced incidence core is acyclic.

A connected instance is a special case, so Lemma A° implies the connected Lemma A used in Theorem 3.13.

The following foothold reduces the problem to cycles with an external neighbour.

Lemma 4.2 (Closed-neighbourhood subcase, A-I)**.** *Suppose the forced core has a cycle F with S-side P and H-side Q ($|P| = |Q| = m$). If $N_H(P) \subseteq Q$, then hereditary tightness is contradicted; hence no such F exists.*

Proof. In $G[H \cup P]$ pick any $q_0 \in Q$ and set $D = Q \setminus \{q_0\}$, $|D| = m - 1$. Any cycle of $G[H \cup P] - D$ lies in $H - D$ (impossible, an induced subgraph of the forest H) or passes through some $s \in P$. The cycle F alternates S- and H-vertices, while the bipartition classes alternate along every edge; hence all S-positions of the even cycle F lie in one class, so P is contained in one bipartition class. Therefore s has no neighbours in P, and P–$(S \setminus P)$ edges are absent from $G[H \cup P]$; with $N_H(s) \subseteq Q$, deleting $Q \setminus \{q_0\}$ leaves each $s \in P$ of degree ≤ 1, hence on no cycle. So D is an FVS of size $m - 1$, giving $\tau(G[H \cup P]) \leq m - 1 < m$, contradicting Lemma 3.3. □

Two saturation facts follow from tightness; both are used to pin down the canonical carriers.

Lemma 4.3 (Tight-set saturation)**.** *For a tight X ($\rho(X) = 2|X|$), every feasible J and every $h \in H$ have $\mathrm{load}_X^J(h) = \min(c(h), \deg_X(h))$, where $\mathrm{load}_X^J(h)$ counts selected X–h incidences.*

Proof. Termwise $\mathrm{load}_X^J(h) \leq \min(c(h), \deg_X(h))$, while

$$\sum_h \mathrm{load}_X^J(h) = 2|X| = \rho(X) = \sum_h \min\big(c(h), \deg_X(h)\big);$$

equal sums of termwise-dominated quantities force equality in every term. □

Lemma 4.4 (Common tight carrier)**.** *For s on a forced cycle, the two forced incidences at s share a single inclusion-minimal tight exposing set A_s, and A_s is the unique minimum tight set containing s.*

Proof. Let X^-, X^+ be inclusion-minimal tight sets exposing the two forced incidences f^-, f^+ at s; these exist by finiteness (Lemma 3.5 gives some tight exposer, and a smallest one exists). By Lemma 3.6, $X^- \cap X^+$ is tight; it contains s and, having coordinatewise degree $\leq$ that of each exposer, still exposes both, so minimality gives $X^- \cap X^+ = X^- = X^+ =: A_s$. For any tight $Y \ni s$, $A_s \cap Y$ is tight, contains s, and exposes a forced incidence at s, so $A_s \cap Y = A_s$, i.e. $A_s \subseteq Y$. □

Lemma 4.5 (Punctured tightness)**.** *Let $B_s = A_s \setminus \{s\}$. Then B_s is tight; the only H-neighbours v of s with $\deg_{A_s}(v) \leq c(v)$ are the two forced ends q^-, q^+; and every other H-neighbour h of s has $\deg_{A_s}(h) \geq c(h) + 1$.*

Proof. Removing s lowers $\min(c(v), \deg_{A_s}(v))$ by one exactly at the neighbours v with $\deg_{A_s}(v) \leq c(v)$; let k be their number. The forced ends q^-, q^+ are two such vertices, so $k \geq 2$; and $\rho(B_s) = \rho(A_s) - k = 2|A_s| - k$, so $\delta(B_s) = 2 - k \geq 0$ gives $k \leq 2$. Hence $k = 2$, $\delta(B_s) = 0$, the two are exactly q^-, q^+, and every other H-neighbour is oversubscribed. □

The first use of the carrier family as a whole gives a separation between a forced cycle and the punctures that support it.

Lemma 4.6 (Carrier–cycle separation (CS))**.** *Suppose the forced core contains the alternating cycle $F = s_1 q_1 s_2 q_2 \cdots s_m q_m s_1$. Put $P = \{s_1, \ldots, s_m\}$, $Q = \{q_1, \ldots, q_m\}$, and $U = \bigcup_{s \in P} B_s$. Then:*

(a) if $s \in P \setminus U$, each of its two forced ends q satisfies $\deg_U(q) \leq c(q) - 1 = 1$;

(b) if consecutive s_i, s_{i+1} both lie in $P \setminus U$, then $\deg_U(q_i) = 0$;

(c) if $P \cap U = \varnothing$, then $\deg_U(q) = 0$ for every $q \in Q$; under the T_F normalization this says $N_S(q) \subseteq P$;

(d) $U \cup R$ is tight for every $R \subseteq P \setminus U$.

Proof. Every B_s and every A_s is tight. Since tight sets are closed under union, both $U = \bigcup_{s \in P} B_s$ and $T = P \cup U = \bigcup_{s \in P} A_s$ are tight.

Fix $s \in P \setminus U$. If $Y \supseteq U$ and $s \notin Y$, then

$$\rho(Y \cup \{s\}) - \rho(Y) = \big|\{h \in N_H(s) : \deg_Y(h) < c(h)\}\big|.$$

For every neighbour h other than the two forced ends, Lemma 4.5 gives $\deg_{A_s}(h) \geq c(h) + 1$, hence $\deg_{B_s}(h) \geq c(h)$; such a neighbour contributes nothing. Thus the marginal is at most 2, with equality precisely when both forced ends remain unsaturated.

Order $P \setminus U = \{p_1, \ldots, p_r\}$ arbitrarily and add these vertices to U in that order. The marginals telescope to

$$\rho(T) - \rho(U) = 2|T| - 2|U| = 2r.$$

Submodularity makes each marginal at most the corresponding single-vertex marginal above, hence at most 2. All are therefore exactly 2. Because the order was arbitrary, every $U \cup R$ is tight, proving (d), and taking one vertex first shows that both forced ends of every $s \in P \setminus U$ have $\deg_U(q) < c(q) = 2$, proving (a).

Now take consecutive $s_i, s_{i+1} \in P \setminus U$ first in the order. The second marginal is 2, so the shared end q_i must still contribute after s_i is added. If $d = \deg_U(q_i)$, this contribution equals $\min(2, d+2) - \min(2, d+1)$, which is 1 only when $d = 0$. This proves (b). If $P \cap U = \varnothing$, every consecutive pair qualifies, so (b) applies to every q_i, proving (c). □

Lemma 4.7 (Single-blocker subcase)**.** *Let h be an external H-neighbour of s with $c(h) = 2$ and $U_h = N_G(h) \cap B_s$. If every $u \in U_h$ has at most one H-neighbour other than h, then $\tau(G[H \cup U_h]) \leq 1 < |U_h|$, a contradiction.*

Proof. By Lemma 4.5, $\deg_{A_s}(h) \geq c(h) + 1 = 3$; since $s \in N(h)$, the remaining neighbours give $|U_h| = |N(h) \cap B_s| \geq 2$. All of U_h share the neighbour h, so lie in one bipartition class and induce no edges among themselves. In $G[H \cup U_h] - h$ each $u \in U_h$ then has degree ≤ 1 and lies on no cycle, and $H - h$ is a forest; so $\{h\}$ is an FVS, giving $\tau(G[H \cup U_h]) \leq 1 < 2 \leq |U_h|$, contradicting hereditary tightness. □

Finally, a normalization that lets one assume S equals the union of carriers.

Lemma 4.8 (T_F normalization)**.** *Suppose a Lemma-A° instance has a forced-core cycle F with S-side P. Put $T_F = \bigcup_{s \in P} A_s$ and restrict to $G' = G[H \cup T_F]$, $S' = T_F$ (same H, edge e, capacities). Then $G' - S' = H$; S' is a minimum FVS of G'; hereditary tightness holds for every $X \subseteq T_F$; the restricted system is feasible; every A_s stays tight and exposes both forced edges at s; and every edge of F remains forced in G'. Hence a minimal counterexample to Lemma A° may be assumed to satisfy $S = T_F$.*

Proof. (i) For every $X \subseteq T_F$, the induced subgraphs $G'[H \cup X]$ and $G[H \cup X]$ are literally the same graph (both are induced on $H \cup X$ in G), so $\tau(G'[H \cup X]) = \tau(G[H \cup X]) = |X|$ verbatim from Lemma 3.3 applied to G; hereditary tightness holds on all $X \subseteq T_F$. (ii) Taking $X = T_F$ gives $\tau(G') = |T_F|$; and T_F is an FVS of G' (deleting it leaves H, a forest), hence a minimum FVS. (iii) The quantity ρ is intrinsic to $(X, H,$ capacities, X-to-H edges), all unchanged in G'; so the Hall inequality $\rho(X) \geq 2|X|$ for all $X \subseteq S$ restricts to all $X \subseteq T_F$, and Lemma 3.4 gives feasibility of the restricted system. (iv) Forcedness is *not* monotone under restriction: removing the demand of S-vertices outside T_F can create new feasible selections. Instead, each F-edge is re-forced via Lemma 3.5: its carrier $A_s \subseteq T_F$ is still tight (same ρ value) and still exposes it (same degrees and capacities), so no feasible selection of G' omits it, and F remains in the forced core of G'. Connectedness may be lost under restriction — which is exactly why Lemma A° is stated without it. □

Warning witnesses. Three certified instances (found and independently recomputed) bracket the false localizations and show why the remaining case is genuinely open. Each eliminates a tempting shortcut. None of them is a counterexample to Lemma A°: each is an instance in which hereditary tightness *fails* at a specific location (so its forced cycle is permitted), and each is used only to refute a proposed rule for *where* that violation must sit — not to exhibit a hereditarily tight instance with a forced cycle, which is exactly what Lemma A° forbids.

- **O-56** (capacity 2, acyclic quotient). Here the hereditary violation sits inside B_s and a *single* external H-vertex blocks it: this is exactly the single-blocker subcase of Lemma 4.7 ($A_s = \{0, 2, 3\}$, $B_s = \{2, 3\}$ tight, forced ends at $\{4, 5\}$, external neighbour 6 with $\deg_{A_s}(6) = 3$; the incidences $2 \sim \{6, 7\}$, $3 \sim \{6, 8\}$). It shows the subcase is nonempty but does not by itself close the general case.
- **O-63** (capacity 2, cyclic quotient). Here every vertex of U_h has incidence-degree 3, the violation is the *entire* $B_s = \{u, a, b\}$, and the blocker $\{h, w\}$ has size 2. This refutes both "every B_s-vertex has incidence-degree 2" and "incidence-degree ≥ 3 forces a private cycle": the blocking set can exceed a single external vertex, and high degree alone does not force a cycle in the forest quotient.

- **O-64** (capacity 1). Explicitly: $S = \{s, t, u\}$; H is the tree $q_1 - r - q_2$ together with $q_1 - y$, $y - h$, $y - w$; fixed edge $e = (h, y)$ with $c(h) = c(y) = 1$; adjacencies $s \sim q_1, q_2, h$, $t \sim q_1, q_2$, $u \sim h, w$. The unique feasible selection sends $s, t \to \{q_1, q_2\}$ and $u \to \{h, w\}$, so u saturates the capacity-1 vertex h, s is pushed off h, and $s - q_1 - t - q_2 - s$ is a forced cycle. Here the carrier puncture $B_s = \{u\}$ is Hall-tight yet $\tau(G[H \cup B_s]) = 1 = |B_s|$: it carries *no* hereditary violation. The first violation sits on the cycle's own S-side $P = \{s, t\}$ ($\tau(G[H \cup P]) = 1 < 2$). This refutes B_s-localization and shows that capacity-1 and capacity-2 external neighbours are separate branches; neither P nor B_s is a universal localizer.

The remaining open target is to derive, from the carrier family $\{A_s : s \in P\}$, their punctures B_s, the external-neighbour structure, and the augmented forest quotient, either a hereditarily violating subset of $P \cup \bigcup_s A_s$ or a feasible exchange avoiding a forced-cycle edge. The witnesses show the violating set need not be P, B_s, or T_F; no single localization is available.

Computationally, Lemma A now holds on every enumerated instance through $n = 14$. The census scope is every minimum FVS with $\tau(G) = \tau(G) \geq 2$, every H-edge; the $\tau \leq 1$ layer is omitted as trivial—an incidence cycle requires two S-vertices, and at $\tau \leq 1$ every feasible selection is acyclic.

n	connected bipartite graphs	instances (G, S, e)	cyclic forced cores
11	25,598	994,607	0
12	212,780	10,970,519	0
13	2,241,730	141,887,501	0
14	31,193,324	2,391,365,507	0

The $n = 11$ layer, previously retained only as an incompletely ledgered run, was re-run reproducibly. Two algorithmically independent engines agree through $n = 11$: the published enumeration–intersection checker and a new flow-based forcedness checker. The flow engine then performed the full runs through $n = 14$. Both engines detect cyclic forced cores on the three warning witnesses. Their frozen source hashes and exact universes are recorded in Section 6.

The sufficient conclusion needed by Theorem 3.13 was also checked directly. For each enumerated instance the checker searched for an acyclic feasible selection, both universally (for every fixed e) and existentially (for some e). At $n = 8, 9, 10, 11, 12, 13$ the respective instance counts were $2,186$, $13,003$, $116,216$, $994,607$, $10,970,519$, and $141,887,501$; both failure counts were zero at every order. The three warning witnesses are positive controls: none admits an acyclic feasible selection.

4.2 Lemma B: cycle-erasing descent

Lemma B asks that a CES toggle exist for every selected cycle that has an avoidable incidence (Lemma A guarantees every selected cycle is of this kind). The natural metric refinement is formulated through residual augmenting paths. For such a cycle C let $d_0(C)$ be the minimum length of a residual augmenting path over the avoidable edges of C, and $d_{\mathrm{CES}}(C)$ the minimum length over CES residual paths.

Open Problem 4.9 (Lemma B)**.** Prove that every selected cycle containing an avoidable incidence admits a CES toggle; equivalently, $d_{\mathrm{CES}}(C) < \infty$ for every such cycle C.

A shortest-path-only formulation of Lemma B (that some globally shortest residual augmenting path is already CES) held exhaustively through $n = 10$ but was falsified at $n = 11$ by a sealed

holdout (six 4-cycle witnesses at which a shortest bad path destroys C but creates one crossing cycle, the cycle-erasing reroute appearing only at length $d_0 + 2$). Two refinements survive those exact witnesses and are supported *in sample only*: the metric bound $d_{\mathrm{CES}}(C) \le d_0(C) + 2$, and a terminal-slide mechanism producing the CES move by a two-edge extension of a shortest path; at $n = 11$ the gap distribution over the 8,202,633 selected cycles was $\{0 : 8{,}202{,}627,\ 2 : 6\}$. These are refinements of an open lemma, not theorems; a prospective $n = 12$ holdout was aborted without a verdict. We record the following conditional step, whose hypotheses the terminal-slide mechanism is meant to supply.

Proposition 4.10 (Terminal-slide criterion)**.** *Let $a = (s, h)$ be an avoidable incidence, Q_0 a simple residual augmenting path from the release ending at an H-terminal t, and $J_0 = \mathrm{toggle}(Q_0)$ feasible, with final added edge (u, t). Suppose (H1) C is not a cycle of J_0; (H2) J_0 has exactly one cycle Z not already present; (H3) (u, t) lies on Z; (H4) (u, h) is an unselected incidence with $u \neq s$ and $\deg_{J_0-(u,t)}(h) < c(h)$; (H5) u and h lie in different components of $J_0 - (u, t)$. Then $J^+ = J_0 - (u, t) + (u, h)$ is feasible, still omits a, and is CES for C.*

Proof. Feasibility is termwise: t loses one incidence; u swaps (u, t) for (u, h), keeping selected degree 2; h gains one incidence but $\deg_{J_0-(u,t)}(h) < c(h)$ by (H4); all other vertices are unchanged. Since $(u, h) \neq a$, the released a stays unselected. For cycles: by (H2)–(H3), Z is the only new cycle and $(u, t) \in Z$, so $J_0 - (u, t)$ has only cycles already present in the pre-toggle selection; adding (u, h) across two different components (H5) creates no cycle. Hence J^+ destroys C and adds no new cycle: it is CES. □

The open content of Lemma B is to prove that, for every positive-gap cycle, a shortest residual path exists whose terminal configuration satisfies (H1)–(H5)—an argument that must invoke residual shortest-path structure and hereditary tightness rather than any single empirical normal form.

5 Toward the general case

We turn to $o > 0$. Two monotonicities are immediate.

Lemma 5.1. *For any vertex set T: $\ell(G) \ge \ell(G - T)$ and $\tau(G - T) \ge \tau(G) - |T|$. In particular for a minimum odd-cycle transversal T with $B = G - T$ bipartite, $\ell(G) \ge \ell(B)$ and $\tau(B) \ge \tau(G) - o$.*

Proof. A spanning linear forest of $G - T$ extends to one of G by adjoining the vertices of T as trivial paths, so $\ell(G) \ge \ell(G - T)$. Deleting $|T|$ vertices lowers the minimum FVS by at most $|T|$, so $\tau(G - T) \ge \tau(G) - |T|$. □

Lemma 5.2 (Edge in the maximum bipartite subgraph)**.** *If G is connected and nontrivial, a maximum induced bipartite subgraph B has an edge; more generally $b(G) \ge \alpha(G) + 1$ where α is the independence number.*

Proof. If B were edgeless then $b(G) = \alpha(G)$. Take a maximum independent set I; since G is connected and nontrivial there is a vertex $v \notin I$ with a neighbour in I, and $I \cup \{v\}$ induces a star, which is bipartite, of size $|I| + 1 > b(G)$, contradicting maximality. □

Theorem 5.3 (Conditional coefficient-two bound)**.** *Assume the bipartite case of Conjecture 1.1 ($\ell(B) \ge 2\tau(B) + 1$ for every connected nontrivial bipartite B). Then for every connected nontrivial graph G,*

$$\ell(G) + 2\,o(G) \ \ge\ 2\,\tau(G) + 1.$$

Proof. Let T be a minimum odd-cycle transversal, $|T| = o$, and $B = G - T$ a maximum induced bipartite subgraph. By Lemma 5.1, $\ell(G) \geq \ell(B)$ and $\tau(B) \geq \tau(G) - o$. Applying the bipartite case componentwise to B (an edgeless component has $\ell = 0 = 2\tau$), $\ell(B) \geq 2\tau(B) + c_e$, where c_e is the number of components of B with an edge. By Lemma 5.2, $c_e \geq 1$. Hence

$$\ell(G) \geq \ell(B) \geq 2\tau(B) + c_e \geq 2(\tau(G) - o) + 1,$$

so $\ell(G) + 2o \geq 2\tau(G) + 1$. □

Theorem 5.3 has coefficient 2 on o; the target has coefficient 1. We isolate a single-vertex property that closes this gap.

Definition 5.4 (SV)**.** Say SV holds if for every connected nonbipartite finite simple graph G there is a vertex t lying in some minimum odd-cycle transversal of G with $p(G) \leq p(G - t)$, equivalently $\ell(G) \geq \ell(G - t) + 1$.

Note t lies in a minimum odd-cycle transversal iff $o(G - t) = o(G) - 1$.

Theorem 5.5 (Single-vertex reduction)**.** *If SV holds and the bipartite case of Conjecture 1.1 holds, then $\ell(G) + o(G) \geq 2\tau(G) + 1$ for every connected nontrivial G; i.e. Conjecture 1.1 holds in full.*

Proof. Strong induction on $|V(G)|$. If G is bipartite, $o(G) = 0$ and the bipartite case gives $\ell(G) \geq 2\tau(G) + 1 = 2\tau(G) + o(G) + 1$. Otherwise G is connected nonbipartite; take t as in SV, so $o(G - t) = o(G) - 1$ and $\ell(G) \geq \ell(G - t) + 1$. Also $\tau(G - t) \geq \tau(G) - 1$. The parameters ℓ, o, τ are additive over connected components, and to each edge-containing component K_i of $G - t$ apply either the induction hypothesis (if nonbipartite, $|K_i| < |G|$) or the bipartite case (if bipartite); both give $\ell(K_i) + o(K_i) \geq 2\tau(K_i) + 1$. Isolated components contribute 0. Summing over the $c_+(G - t)$ edge-containing components,

$$\ell(G - t) + o(G - t) \;\geq\; 2\tau(G - t) + c_+(G - t).$$

Finally $c_+(G - t) \geq 1$: G has an odd cycle C; if $t \notin C$ then C survives, and if $t \in C$ then $C - t$ is a path with an edge, so $G - t$ has an edge. Combining,

$$\ell(G)+o(G) \;\geq\; (\ell(G-t)+1)+(o(G-t)+1) \;\geq\; (2\tau(G-t)+c_+)+2 \;\geq\; 2(\tau(G)-1)+1+2 = 2\tau(G)+1.$$

□

The implication in Theorem 5.5 remains valid, but its hypothesis is false. The smallest witness found has graph6 string `J???FAWdVD?` and edge set

$$\{07, 08, 09, 0\,10, 17, 1\,10, 27, 2\,10, 38, 39, 48, 59, 6\,10, 79, 8\,10\}.$$

It has $f = 9$, $b = 10$, $p = 3$, and $o = 1$. The triangles 0–7–9–0 and 0–8–10–0 meet only at 0, while $G - 0$ is bipartite. Thus $\{0\}$ is the unique minimum odd-cycle transversal, but $p(G - 0) = 2 < 3 = p(G)$, so the sole eligible vertex fails SV. A fresh exhaustive check found no failure among all 272,209 connected nonbipartite graphs on at most 9 vertices or all 11,712,539 at order 10. Hence the minimum order of an SV counterexample is 11. SV was only a sufficient route, so this does not refute Conjecture 1.1.

The surviving lifting route can be stated directly. For a minimum odd-cycle transversal T, put $B = G - T$.

Definition 5.6 (Lifting statements)**.** LIFT-A says that for some minimum T,

$$\ell(G) \geq 2\tau(G-T) + |T| + 1;$$

LIFT-A-ALL requires this for every minimum T. LIFT-P says that for some minimum T, $p(G) \leq p(G-T)$, and LIFT-P-ALL again requires every minimum T.

LIFT-P is false on the SV witness: its minimum transversal is uniquely $T = \{0\}$ and $p(G) = 3 > 2 = p(G-T)$. Thus the path-cover lift dies with SV. LIFT-A and even LIFT-A-ALL survive; their slack $\ell(G) - [2\tau(G-T) + |T| + 1]$ is 4 on each of the three certified SV witnesses. Moreover, LIFT-A together with the bipartite case implies the full conjecture, since $\tau(G-T) \geq \tau(G) - |T|$ gives

$$\ell(G) \geq 2\tau(G-T) + |T| + 1 \geq 2\tau(G) - |T| + 1.$$

The following two lemmas prove LIFT-A on a structured class.

Lemma 5.7 (L1)**.** *If B is bipartite on n_B vertices and has $c_e(B)$ edge-containing components, then*

$$f(B) \geq \left\lceil \frac{n_B}{2} \right\rceil + c_e(B), \qquad \tau(B) \leq \left\lfloor \frac{n_B}{2} \right\rfloor - c_e(B).$$

Proof. In each edge-containing component, take the larger bipartition class and one vertex of the other class. The induced subgraph is a star together with isolated vertices, hence a forest, and has at least $\lceil n_i/2 \rceil + 1$ vertices. Take every isolated vertex of B as well. The componentwise union is an induced forest, and summing the ceilings gives the stated bound. The equivalent FVS bound follows from $\tau(B) = n_B - f(B)$. □

Lemma 5.8 (L2)**.** *Let G be connected and nonbipartite. If some minimum odd-cycle transversal T satisfies*

$$p(G) \leq 2c_e(G-T) - 1,$$

then LIFT-A holds for G with that T.

Proof. Write $B = G - T$ and $c_e = c_e(B)$. Lemma 5.7 gives

$$2\tau(B) + |T| + 1 \leq 2\left\lfloor \frac{|B|}{2} \right\rfloor - 2c_e + |T| + 1 \leq |V(G)| + 1 - 2c_e \leq |V(G)| - p(G) = \ell(G).$$

□

In particular, every traceable connected nonbipartite graph satisfies LIFT-A: $p(G) = 1$, while Lemma 5.2 ensures $c_e(G-T) \geq 1$. Consequently the conjecture holds on this class. Exhaustively over connected nonbipartite graphs with $3 \leq n \leq 8$, the tightness cross-tabulation is:

best LIFT-A slack	0	1	2	3	4	5
L2-covered	217	666	2,773	3,978	3,255	281
L2-uncovered	0	43	249	199	198	0

Thus every LIFT-A-tight graph in this universe lies in the proved class; every uncovered graph has slack at least one. The computational evidence for the surviving lift is also exact through order 10: LIFT-A and LIFT-A-ALL have zero failures on every connected graph with $3 \leq n \leq 10$. At $n = 10$ this is all $11,716,571$ connected graphs, and $52,303$ nonbipartite graphs are tight. An independent full census reproduced both zero failure counts and the same tight list. A preregistered $n = 11$ holdout comprising residue classes $1, 2, 3 \pmod{1024}$ contained $2,051,196$ connected graphs ($2,045,943$ nonbipartite): LIFT-A and LIFT-A-ALL again had zero failures, while SV and LIFT-P each had 19.

6 Computational verification

Methodology. All computational claims here come from a preregistered protocol. A single exact checker `phase0.c` computes f (maximum induced forest, by a subset scan with union-find acyclicity test), b (maximum induced bipartite subgraph, by a subset scan with a 2-colouring test), and p (minimum path cover $= n-$ maximum spanning linear forest, by a Hamiltonicity/DP bitmask) exactly for $n \leq 11$. The checker source was frozen and SHA-256 hashed before any run; a gold suite (invariants on K_{10}, $K_{5,5}$, $K_{1,9}$, complete multipartite families, C_{10}, and others) and a 995-graph regression over $n \leq 7$ passed before launch. Graph enumeration used `nauty geng -c` [4]. Every low-slack graph was re-checked by a second, independently coded engine (`phase0_verify2.py`, a pure-Python bitmask implementation sharing no code with the C checker). Any bound violation would halt the run with the graph and witness sets; none occurred.

Direct verification of Conjecture 1.1.

- $2 \leq n \leq 10$: every connected graph was enumerated, with 0 violations at every order. The connected-graph counts matched the known values exactly: $n = 2$ through 7 give $1, 2, 6, 21, 112, 853$ (total 995, the regression layer); $n = 8$ gives 11,117; $n = 9$ gives 261,080; $n = 10$ gives 11,716,571. The layers $n = 8, 9$ were run under the same re-frozen checker as $n = 11$ (re-golded before use), so the whole range sits under one dual-engine protocol. At $n = 10$ the minimum slack is 0, with 115,154 low-slack graphs (slack ≤ 1) of which 4,960 meet the bound with equality ($2f = p + b + 1$); all low-slack graphs of every order $n \leq 10$, plus a 10,005-graph sample at $n = 10$, were re-verified by the second engine with 0 violations. (Checker `phase0.c`, freeze hashes `9f5cc322` for the original $n \leq 10$ runs and `bc7f614d` for the re-frozen $n = 8, 9$ and $n = 11$ runs.)
- $n = 11$: all 1,006,700,565 connected graphs on 11 vertices were enumerated, sharded into 192 residue classes whose counts sum to exactly 1,006,700,565; there were 0 violations, minimum slack 0, and 3,853,461 low-slack graphs, of which 96,949 meet the bound with equality (and 3,756,512 have slack 1). All 3,853,461 low-slack graphs plus a 9,968-graph sample — 3,863,429 graphs in all — were re-verified by the second engine, with 0 violations. (Checker `phase0.c`, re-freeze hash `bc7f614d`; wall time 18.1 hours on an 8-core machine.)

Thus Conjecture 1.1 holds, with no exceptions, for every connected graph on at most 11 vertices.

SV and the lifting census. For each connected nonbipartite G, the eligible set $\{t : o(G - t) = o(G) - 1\}$ was computed and tested directly. There were no failures among the $272,209$ graphs through $n = 9$ or the $11,712,539$ graphs at $n = 10$. In the preregistered class 3 (mod 1024) at $n = 11$, both the lifting engine and the published frozen SV checker found exactly 19 failures among $363,207$ connected nonbipartite graphs; the explicit witnesses were then checked independently from the definitions. The larger three-class lifting holdout and the completed release census are summarized below.

universe	connected	nonbipartite	LIFT-A fail	LIFT-A-ALL fail	tight
$3 \leq n \leq 9$ (full)	273,191	272,209	0	0	2,494 at $n = 9$
$n = 10$ (full)	11,716,571	11,712,539	0	0	52,303
$n = 11$, classes $1, 2, 3/1024$	2,051,196	2,045,943	0	0	5,093

Independent computational protocols. The attempt fixed its statements, controls, and source hashes before its sweeps. VERIFY-1 then used an adversarial re-verification protocol: the independent checkers and proof derivations were written from the paper's definitions before the attempt code was read. The hashes of the exact sources used for the reported runs are:

engine	SHA-256
published `verification/phase0.c`	`17db663be9cd4d1e9adb1524b2717fb90c379fc53ba56066a5d3bcf275c8dac6`
attempt `work/lift.c`	`c0cc2dd315144a0c9c64c875234e193fd8f700e01949ea6668964220a4d01c5e`
VERIFY-1 lifting engine	`af5e7cbf5b886357fba68f1ecca01ef22caf925c8b0fe6f75f2295495bdc95cd`
attempt `work/fcore.c`	`598bb06f358e21ae47f0e1443029ea3c3b07c8bd2d6b2d5eb8189319ab463a7c`
VERIFY-1 forced-core engine	`25ce6906a24452d2959314f1ef3c3425259fe3bd76f64207c02cbe97404dd5ed`
attempt `work/f2.c`	`e528a4ae58ece0489c50a11e6f4a3ff229e3a24383ec856d3a0877994549b458`
VERIFY-1 F2 engine	`fd55562bb3dbbe6531f8da8171ea2a1d9c820f0d1a047719d67cb2d5c63c4171`

The v2 copies of `work/fcore.c` and `work/f2.c` change only their census-scope comments; the executed logic is unchanged. Their respective SHA-256 values are

`989f5a686ba33e5da95e9c2011cae847bd4570430b13e7244b5ae4d9b4b54bf5`
`40798ae0823628d230c9e226bead849ed01f223f482bae99de1b8f6b0c6ef46f`

The attempt's forced-core engine uses the residual-flow characterization; VERIFY-1 instead intersects independently enumerated feasible selections. Its F2 checker is likewise independent. The resulting universes agree exactly:

claim	orders	instances checked	failures
SV minimality	$3 \le n \le 10$	11,984,748 nonbipartite graphs	0
SV witness class	$n = 11$, 3/1024	363,207 nonbipartite graphs	19
Lemma A	$11 \le n \le 14$	2,545,218,134 (G, S, e)	0 cyclic cores
F2-ALL and F2-SOME	$8 \le n \le 13$	153,984,032 (G, S, e)	0 / 0
LIFT-A and LIFT-A-ALL	$3 \le n \le 10$	11,984,748 nonbipartite graphs	0 / 0

For the Lemma-A and F2 censuses, the precise scope is every minimum FVS with $\tau \ge 2$, every H-edge; the $\tau \le 1$ layer is omitted as trivial—an incidence cycle requires two S-vertices, and at $\tau \le 1$ every feasible selection is acyclic. Positive controls were run first: all three warning witnesses have cyclic forced cores and none has an acyclic feasible selection.

As enumeration anchors, independent `geng -c -b` counts matched OEIS A005142 [5] at every new order:

n	connected bipartite graphs	Lemma-A instances	cyclic cores
11	25,598	994,607	0
12	212,780	10,970,519	0
13	2,241,730	141,887,501	0
14	31,193,324	2,391,365,507	0

The independently regenerated tight-graph lists have $2,494$ records at $n = 9$ and $52,303$ at $n = 10$. Their respective SHA-256 values are

`6078480935bf84787a02b5ee580eb13e039583419929013916379f97c3cd5abc`
`9fe71fb1c6675b06f59901060898304509b5179358be9967188cd88a25566a69`

Status of the lemma-level runs. Lemma A has no cyclic forced core in the full census through $n = 14$, and the stronger direct selection property F2-ALL has no failure through $n = 13$. Lemma B's shortest-only form was verified exhaustively through $n \le 10$ and *falsified* at $n = 11$ by a sealed holdout (six witnesses), after which the surviving metric and terminal-slide refinements were supported only in sample (Section 4); a prospective $n = 12$ holdout was aborted without a verdict.

7 Related work

The nearest neighbour is the work of Lavrov and Vandenbussche [1]. They do not characterize path covers; they *conjecture* a Hall-type sufficient condition under which one side X of a bipartite graph can be covered by at most $\Lambda\text{-def}(G)$ vertex-disjoint paths, and they *prove* the conjecture for forests, for graphs of maximum degree 3, and for regular graphs of high girth. Their Λ-deficiency and the demand-2 deficiency δ_2 of our incidence model (Section 3.1) are related. Taking the model with *all capacities* 2 (no capacity-1 pair), write, for a vertex set S, $\mathrm{supp}(S)$ for the number of H-vertices with at least one neighbour in S, and $\Lambda(S)$ for the number with at least two; then the two deficiencies satisfy

$$\delta_2(S) \;+\; \Lambda\text{-def}(S) \;=\; \mathrm{supp}(S) \;-\; |S|.$$

[1] The obstacle to transferring their machinery directly is that Λ-def is neither submodular nor supermodular, so the tight-set lattice (Lemma 3.6) that drives our carrier analysis has no counterpart on their side. Separately, the parameters $f = n - \tau$ and the decycling number τ sit in the large induced-forest / feedback-vertex-set literature, and $b = n - o$ with o in the odd-cycle-transversal literature; the novelty of the deficiency form (Theorem 2.1) is that it couples all three of τ, o, and the linear-forest parameter ℓ in a single inequality.

8 Data availability

The frozen checkers, independent VERIFY-1 engines, per-shard enumeration logs, SV certificates, and hashed LIFT-A-tight graph lists are available in the repository at `https://github.com/cozuya/wowii-conjecture-40`, under release tag `v2.0`, and in the versioned archive at `https://doi.org/10.5281/zenodo.21778700`.

Changes from version 1

Version 2 strengthens the record; it is not an erratum, and version 1 contains no false statement. SV, posed open and verified only through order 10 in version 1, is now resolved as false with minimum counterexamples at order 11. The Lemma-A census is reproducible and extended through order 14, F2 is verified directly through order 13, and the surviving lifting inequality is proved on the L2 class and independently re-verified through order 10.

Acknowledgments

The mathematics and the computations in this paper were developed with AI assistance (Claude models); all results were independently cross-checked computationally.

References

[1] M. Lavrov and J. Vandenbussche, *A Hall-type condition for path covers in bipartite graphs*, Electronic Journal of Combinatorics **31**(3) (2024), #P3.1. doi:10.37236/12462.

[1] This identity is stated here as the bridge between the two deficiency measures; it is not derived from the proof documents of this project and we do not claim it as one of the results verified here.

[2] E. DeLaviña, *Written on the Wall II* (Graffiti.pc conjectures), Conjecture 40. Maintained open-problems list, http://cms.uhd.edu/faculty/delavinae/research/wowII/open.html (last updated 2026-07-23; accessed 2026-08-03).

[3] Google DeepMind, *formal-conjectures* (Lean), `GraphConjecture40`. https://github.com/google-deepmind/formal-conjectures, file `FormalConjectures/WrittenOnTheWallII/GraphConjecture40.lean` (accessed 2026-08-03).

[4] B. D. McKay and A. Piperno, *Practical graph isomorphism, II*, J. Symbolic Comput. **60** (2014), 94–112. doi:10.1016/j.jsc.2013.09.003.

[5] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, Sequence A005142, number of connected bipartite graphs with n vertices, https://oeis.org/A005142 (accessed 2026-08-03).